Q.1 Numerically approximate the solution curve of the following initial value problem using ode45 command. Also plot the approximated solution curve: $\frac{dy}{dt} = \frac{y}{t^2} + \frac{2}{t}$, $y(0) = 1$

Q.2 Create the following two matrices in MATLAB

$$A = \begin{bmatrix} \frac{3}{11} & \frac{2}{5} & \frac{4}{7} \\ \frac{2}{3} & \frac{4}{5} & \frac{6}{5} \\ \frac{7}{9} & \frac{8}{6} & \frac{9}{4} \end{bmatrix}, \quad B = \begin{bmatrix} 1/3 & 8/9 & 4 \\ 2/9 & 3/7 & 9/6 \\ 6 & 7/4 & \frac{5}{7} \end{bmatrix}$$

i Find the inverse of Matrix A directly using the command of MATLAB.

ii Find A*B using MATLAB.

iii Find the Sum of elements of 1000 row of matrix A directly using the command of MATLAB.

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

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Q no.3 For the following matrix using the MATLAB commands.

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 9 & 5 & 6 \\ 1 & 1 & 2 \end{bmatrix}$$

i) Compare the eign values and eigen vectors for A.

ii) Find the reduced echelon form of the matrix A.

Q.4 write the MATLAB code for approximating the following integral using Simpson's 1/3 rule with n = 10

$$\int_{1}^{2} (\cos(x) - \ln(x) + e^{x})\, dx$$

Q no.5 write a function which generates the Polynomials from the following recurrence relation.

$$L_{i+2}(z) = \frac{i+3}{i+2} L_i(z) + \frac{i}{i+2} L_{i+1}(z)$$

where $L_1(z) = z$ and $L_2(z) = z^2$

Function must take the value of i as input and generates the polynomials.

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